

Distributions & Statistics Notes

Probability Distributions & Descriptive Statistics — Notes with Solved Examples

Covers: Binomial, Poisson, Geometric, Uniform, Exponential, Normal/Gaussian, Chi-Square, Gamma, Beta distributions + Mean, Median, Mode, Correlation & Covariance — each with theory, formulas, and a fully solved numerical example.

PART A — PROBABILITY DISTRIBUTIONS

1. Binomial Distribution

Definition: Models the number of successes in **n independent trials**, each with only two outcomes (success/failure) and a **fixed** probability of success **p**.

Property	Formula
PMF	$P(X=k) = C(n,k) \cdot p^k \cdot (1-p)^{n-k}$
Parameters	n = number of trials, p = probability of success
Mean	$\mu = n \cdot p$
Variance	$\sigma^2 = n \cdot p \cdot (1-p)$
Mode	$\lfloor (n+1)p \rfloor$
Median	$\lfloor n \cdot p \rfloor$ (approx.)

✓ **Solved Example** A fair coin (p = 0.5) is tossed **n = 10** times. Find the probability of getting **exactly 6 heads**.

```
P(X=6) = C(10,6) * (0.5)^6 * (0.5)^4
        = 210 * 0.5^10
        = 210 / 1024
        = 0.2051 (approx. 20.5%)

Mean    = n * p      = 10 * 0.5 = 5
Variance = n * p * (1-p) = 10 * 0.5 * 0.5 = 2.5
```

➔ **Interpretation:** On average, you expect 5 heads out of 10 tosses, and there is a approx. 20.5% chance of getting exactly 6 heads.

2. Poisson Distribution

Definition: Models the number of events occurring in a **fixed interval of time/space**, when events happen independently at a constant average rate **λ**.

Property	Formula
PMF	$P(X=k) = (e^{-\lambda} \cdot \lambda^k) / k!$
Parameter	λ = average rate of occurrence
Mean	$\mu = \lambda$
Variance	$\sigma^2 = \lambda$ (mean = variance, unique property)
Mode	$\lfloor \lambda \rfloor$

✓ **Solved Example** A call center receives an average of **λ = 4 calls per hour**. Find the probability of receiving **exactly 2 calls** in the next hour.

```
P(X=2) = (e^-4 * 4^2) / 2!
        = (0.018316 * 16) / 2
        = 0.29306 / 2
        = 0.1465 (approx. 14.65%)

Mean    = λ = 4
Variance = λ = 4
```

➔ **Interpretation:** There is about a 14.65% chance that exactly 2 calls arrive in the next hour; on average 4 calls are expected.

3. Geometric Distribution

Definition: Models the number of trials needed to get the **first success** in repeated independent Bernoulli trials.

Property	Formula
PMF	$P(X=k) = (1-p)^{k-1} \cdot p$
Parameter	p = probability of success
Mean	$\mu = 1/p$
Variance	$\sigma^2 = (1-p)/p^2$
Mode	1 (always)

✓ **Solved Example** A salesperson closes a deal with probability **p = 0.3** on each call. Find the probability that the **first success (sale) happens on the 4th call**.

$$\begin{aligned}
 P(X=4) &= (1-0.3)^4 \times 0.3 \\
 &= (0.7)^4 \times 0.3 \\
 &= 0.343 \times 0.3 \\
 &= 0.1029 \quad (\text{approx. } 10.29\%)
 \end{aligned}$$

$$\begin{aligned}
 \text{Mean} &= 1/p &= 1/0.3 &= 3.33 \text{ calls} \\
 \text{Variance} &= (1-p)/p^2 &= 0.7/0.09 &= 7.78
 \end{aligned}$$

➔ **Interpretation:** There's a approx. 10.3% chance the first sale happens exactly on the 4th call; on average, it takes about 3.33 calls to get the first sale.

4. Uniform Distribution (Continuous)

Definition: All values in the interval **[a, b]** are **equally likely** — a flat, constant probability density.

Property	Formula
PDF	$f(x) = 1/(b-a)$, for $a \leq x \leq b$
Mean	$\mu = (a+b)/2$
Median	$(a+b)/2$ (= Mean, always)
Mode	Any value (undefined / flat)
Variance	$\sigma^2 = (b-a)^2/12$

✓ **Solved Example** A bus arrival time X is uniformly distributed between **a = 2** and **b = 10 minutes**. Find $P(4 < X < 8)$ and the mean waiting time.

$$\begin{aligned}
 f(x) &= 1/(10-2) = 1/8 \\
 P(4 < X < 8) &= (8-4) \times (1/8) = 4/8 = 0.5 \quad (50\%) \\
 \text{Mean} &= (2+10)/2 &= 6 \text{ minutes} \\
 \text{Variance} &= (10-2)^2/12 &= 64/12 = 5.33
 \end{aligned}$$

➔ **Interpretation:** There's a 50% chance the bus arrives between minute 4 and 8; the expected (average) waiting time is 6 minutes.

5. Exponential Distribution

Definition: Models the **waiting time between events** in a Poisson process (continuous version of the geometric distribution).

Property	Formula
PDF	$f(x) = \lambda \cdot e^{-(\lambda x)}$, $x \geq 0$
Parameter	λ = rate of occurrence
Mean	$\mu = 1/\lambda$
Median	$\ln(2)/\lambda$
Mode	0 (always)
Variance	$\sigma^2 = 1/\lambda^2$
Special property	Memoryless: $P(X>s+t \mid X>s) = P(X>t)$

✓ **Solved Example** Customers arrive at a shop at rate **$\lambda = 0.5$ per minute** (mean gap = 2 minutes). Find the probability that the **next customer takes more than 3 minutes** to arrive.

$$\begin{aligned}
 P(X>3) &= e^{-(\lambda x)} = e^{-(0.5 \times 3)} = e^{-1.5} = 0.2231 \quad (\text{approx. } 22.3\%) \\
 \text{Mean} &= 1/\lambda &= 1/0.5 &= 2 \text{ minutes} \\
 \text{Median} &= \ln(2)/\lambda &= 0.693/0.5 &= 1.386 \text{ minutes}
 \end{aligned}$$

➔ **Interpretation:** There's a approx. 22.3% chance you wait more than 3 minutes for the next customer; on average, customers arrive every 2 minutes.

6. Normal / Gaussian Distribution

(Normal and Gaussian are the **same** distribution — two names for it.)

Definition: A symmetric, bell-shaped continuous distribution — the most common model for natural measurements (heights, marks, errors).

Property	Formula
PDF	$f(x) = (1/(\sigma\sqrt{2\pi})) \cdot e^{-(x-\mu)^2/2\sigma^2}$
Parameters	μ = mean, σ = standard deviation
Mean = Median = Mode	all equal to μ (perfect symmetry)
Variance	σ^2
Empirical Rule	68% within $\mu \pm \sigma$, 95% within $\mu \pm 2\sigma$, 99.7% within $\mu \pm 3\sigma$

✓ **Solved Example** Student marks are normally distributed with **$\mu = 70$, $\sigma = 10$** . Find $P(60 < X < 80)$.

$$\begin{aligned}
 z_1 &= (60-70)/10 = -1 \\
 z_2 &= (80-70)/10 = +1 \\
 P(-1 < Z < 1) &= 0.6826 \quad (\text{Empirical Rule: } 68\% \text{ of data}) \\
 \text{So } P(60 < X < 80) &= 0.6826 \quad (\text{approx. } 68.3\%)
 \end{aligned}$$

➔ **Interpretation:** About 68% of students score between 60 and 80 marks — this is exactly the “68” part of the 68-95-99.7 empirical rule ($\mu \pm 1\sigma$).

7. Chi-Square (χ^2) Distribution

Definition: Sum of squares of k independent standard normal variables; widely used to test **independence between categorical variables** or **goodness-of-fit**.

Property	Formula
Parameter	k = degrees of freedom (df)
Mean	$\mu = k$
Variance	$\sigma^2 = 2k$
Mode	$k-2$ (for $k \geq 2$)
Shape	Right-skewed; approaches Normal as k increases
Test statistic	$\chi^2 = \sum (O - E)^2 / E$

✓ **Solved Example (Goodness-of-fit)** A die is rolled **60 times**. Expected frequency (if fair) for each face = $60/6 = 10$. Observed frequencies: 8, 12, 10, 7, 14, 9. Test if the die is fair at $df = 5$.

$$\begin{aligned}\chi^2 &= \sum (O-E)^2/E \\ &= (8-10)^2/10 + (12-10)^2/10 + (10-10)^2/10 + (7-10)^2/10 + (14-10)^2/10 + (9-10)^2/10 \\ &= 4/10 + 4/10 + 0/10 + 9/10 + 16/10 + 1/10 \\ &= 0.4 + 0.4 + 0 + 0.9 + 1.6 + 0.1 \\ &= 3.4\end{aligned}$$

Critical value ($df=5, \alpha=0.05$) = 11.07

Since calculated χ^2 (3.4) < critical value (11.07) → Fail to reject H_0

➔ **Interpretation:** The observed roll counts are consistent with a fair die (no significant deviation) — used exactly like `scipy.stats.chi2_contingency()` on a `pd.crosstab()` table in EDA.

8. Gamma Distribution

Definition: Generalization of the exponential distribution — models the **waiting time until the k -th event** occurs in a Poisson process.

Property	Formula
PDF	$f(x) = (\lambda^k \cdot x^{k-1} \cdot e^{-(\lambda x)}) / \Gamma(k)$
Parameters	k = shape (number of events), λ = rate
Mean	$\mu = k/\lambda$
Variance	$\sigma^2 = k/\lambda^2$
Mode	$(k-1)/\lambda$, for $k \geq 1$

✓ **Solved Example** Buses arrive at rate $\lambda = 2$ per hour. Find the expected time and variance for the **3rd bus** to arrive ($k = 3$).

$$\begin{aligned}\text{Mean} &= k/\lambda = 3/2 = 1.5 \text{ hours} \\ \text{Variance} &= k/\lambda^2 = 3/4 = 0.75 \\ \text{SD} &= \sqrt{0.75} = 0.866 \text{ hours}\end{aligned}$$

➔ **Interpretation:** You’d expect to wait 1.5 hours (90 minutes) on average for the 3rd bus to arrive, with a standard deviation of about 52 minutes.

9. Beta Distribution

Definition: Models a random variable **bounded between 0 and 1** — ideal for probabilities, proportions, and rates (e.g., conversion rate).

Property	Formula
PDF	$f(x) = x^{\alpha-1}(1-x)^{\beta-1} / B(\alpha,\beta)$
Parameters	α, β = shape parameters
Mean	$\mu = \alpha/(\alpha+\beta)$
Variance	$\sigma^2 = \alpha\beta / [(\alpha+\beta)^2(\alpha+\beta+1)]$
Mode	$(\alpha-1)/(\alpha+\beta-2)$, for $\alpha, \beta > 1$

✓ **Solved Example** A website’s click-through rate is modeled as Beta with $\alpha = 2$ (successes) and $\beta = 3$ (failures) based on past data. Find the mean, mode, and variance of the conversion rate.

$$\begin{aligned}\text{Mean} &= \alpha/(\alpha+\beta) = 2/5 = 0.40 \text{ (40\%)} \\ \text{Mode} &= (\alpha-1)/(\alpha+\beta-2) = 1/3 = 0.333 \text{ (33.3\%)} \\ \text{Variance} &= \alpha\beta / [(\alpha+\beta)^2(\alpha+\beta+1)] = 6/(25 \times 6) = 6/150 = 0.04 \\ \text{SD} &= \sqrt{0.04} = 0.20\end{aligned}$$

➔ **Interpretation:** The expected click-through rate is 40%, with the most likely value (mode) around 33.3% — Beta is the standard “prior belief” distribution for a probability in Bayesian A/B testing.

✦ Quick Comparison Table

Distribution	Type	Parameters	Mean	Variance	Typical Use
Binomial	Discrete	n, p	np	$np(1-p)$	Fixed # of trials, success/fail
Poisson	Discrete	λ	λ	λ	Events per time/space interval

Geometric	Discrete	p	1/p	(1-p)/p²	Trials until 1st success
Uniform	Continuous	a, b	(a+b)/2	(b-a)²/12	Equally-likely range
Exponential	Continuous	λ	1/λ	1/λ²	Time between events
Normal/Gaussian	Continuous	μ, σ	μ	σ²	Symmetric natural data
Chi-Square	Continuous	k	k	2k	Categorical independence test
Gamma	Continuous	k, λ	k/λ	k/λ²	Time until k-th event
Beta	Continuous	α, β	α/(α+β)	αβ/[(α+β)²(α+β+1)]	Probabilities/proportions

PART B — MEAN, MEDIAN, MODE

Measure	Formula	Sensitive to Outliers?	Best For
Mean	$\mu = \sum x / n$	Yes (highly)	Symmetric data (Normal)
Median	Middle value after sorting	No (robust)	Skewed data (income, prices)
Mode	Most frequently occurring value	No	Categorical data

Skewness Rule (relationship between the three):

- Symmetric distribution → Mean = Median = Mode
- Right-skewed (long tail right) → Mean > Median > Mode
- Left-skewed (long tail left) → Mean < Median < Mode

✓ **Solved Example** Marks of 10 students: 45, 50, 55, 55, 60, 65, 70, 75, 80, 95

Mean:
Sum = 45+50+55+55+60+65+70+75+80+95 = 650
Mean = 650 / 10 = 65

Median (n=10, even → average of 5th & 6th sorted values):
Sorted: 45,50,55,55,60,65,70,75,80,95
5th value = 60, 6th value = 65
Median = (60+65)/2 = 62.5

Mode:
55 appears twice (most frequent) → Mode = 55

→ **Interpretation:** Mean (65) > Median (62.5) > Mode (55) → data is **slightly right-skewed** (the high value 95 pulls the mean upward). In pandas: `df['Marks'].mean()` , `df['Marks'].median()` , `df['Marks'].mode()[0]` .

PART C — CORRELATION & COVARIANCE

Covariance

Measures whether two variables move **together (same direction)** or **oppositely**.

$$Cov(X,Y) = \frac{\sum [(x_i - \bar{x})(y_i - \bar{y})]}{(n - 1)}$$

- Range: **-∞ to +∞** (unbounded — depends on the units of X and Y)
- Positive → move in the same direction | Negative → move in opposite directions
- Cannot compare magnitude across different variable pairs (units differ)

Correlation

Standardized version of covariance — tells **direction AND strength** on a fixed scale.

$$r = Cov(X,Y) / (\sigma_x \cdot \sigma_y)$$

- Range: **always -1 to +1**
- r = +1 → perfect positive linear relationship
- r = -1 → perfect negative linear relationship
- r = 0 → no linear relationship

	Covariance	Correlation
Range	-∞ to +∞	-1 to +1
Units	Product of original units	Unit-free
Comparable across variable pairs?	No	Yes
Tells strength of relationship?	No (direction only)	Yes

✓ **Solved Example** Hours studied (X) vs Marks scored (Y) for 5 students:

Student	Hours (X)	Marks (Y)
1	2	50
2	4	60
3	6	65
4	8	80
5	10	90

Step 1 – Means:
 $\bar{x} = (2+4+6+8+10)/5 = 6$
 $\bar{y} = (50+60+65+80+90)/5 = 69$

Step 2 – Deviations:
(X- $\bar{x}$): -4, -2, 0, 2, 4
(Y- $\bar{y}$): -19, -9, -4, 11, 21

Step 3 – Products (X- $\bar{x}$)(Y- $\bar{y}$):
(-4×-19)=76, (-2×-9)=18, (0×-4)=0, (2×11)=22, (4×21)=84
Sum = 76+18+0+22+84 = 200

Step 4 – Covariance:
Cov(X,Y) = 200 / (5-1) = 200/4 = 50

Step 5 – Standard deviations:
 $\Sigma (X-\bar{x})^2 = 16+4+0+4+16 = 40 \rightarrow \sigma_x = \sqrt{(40/4)} = \sqrt{10} = 3.162$
 $\Sigma (Y-\bar{y})^2 = 361+81+16+121+441 = 1020 \rightarrow \sigma_y = \sqrt{(1020/4)} = \sqrt{255} = 15.969$

Step 6 – Correlation:
 $r = \text{Cov}(X,Y) / (\sigma_x \times \sigma_y) = 50 / (3.162 \times 15.969) = 50 / 50.49 = 0.990$

➔ **Interpretation:** r is approx. **0.99** ➔ an extremely strong positive linear relationship: more study hours are almost perfectly associated with higher marks. In pandas:
`df[['Hours','Marks']].cov()` and `df[['Hours','Marks']].corr()` give the same numbers directly.

 Exam-Ready One-Line Summary Table

Term	One-line meaning
Binomial	Count of successes in n fixed trials
Poisson	Count of events in a fixed interval, rate λ
Geometric	Trials until 1st success
Uniform	All values in a range equally likely
Exponential	Time until next event (memoryless)
Normal/Gaussian	Symmetric bell curve, mean=median=mode
Chi-Square	Test independence/goodness-of-fit for categorical data
Gamma	Time until k-th event (generalized exponential)
Beta	Models probabilities/proportions in [0,1]
Mean	Average — sensitive to outliers
Median	Middle value — robust to outliers
Mode	Most frequent value
Covariance	Direction of relationship, unbounded, unit-dependent
Correlation	Direction + strength, always between -1 and +1